

Problem 13. Find all ordered pairs (a, b) of positive integers for which the numbers

$$\frac{a^3b-1}{a+1} \quad \text{and} \quad \frac{b^3a+1}{b-1}$$

are both integers.

Solution. As $a^3b-1 = b(a^3+1) - (b+1)$ and $a+1 \mid a^3+1$, we have $a+1 \mid b+1$. As $b^3a+1 = a(b^3-1) + (a+1)$ and $b-1 \mid b^3-1$, we have $b-1 \mid a+1$. So $b-1 \mid b+1$ and hence $b-1 \mid 2$.

- If $b=2$, then $a+1 \mid b+1=3$ gives $a=2$. Hence $(a, b) = (2, 2)$ is the only solution in this case.
- If $b=3$, then $a+1 \mid b+1=4$ gives $a=1$ or $a=3$. Hence $(a, b) = (1, 3)$ and $(3, 3)$ are the only solutions in this case.

To summarize, $(a, b) = (1, 3), (2, 2)$ and $(3, 3)$ are the only solutions. $\square$

Problem 14. Let Γ be the circumcircle of an acute-angled triangle ABC . The perpendicular to AB from C meets AB at D and Γ again at E . The bisector of angle C meets AB at F and Γ again at G . The line GD meets Γ again at H and the line HF meets Γ again at I . Prove that $AI = EB$.

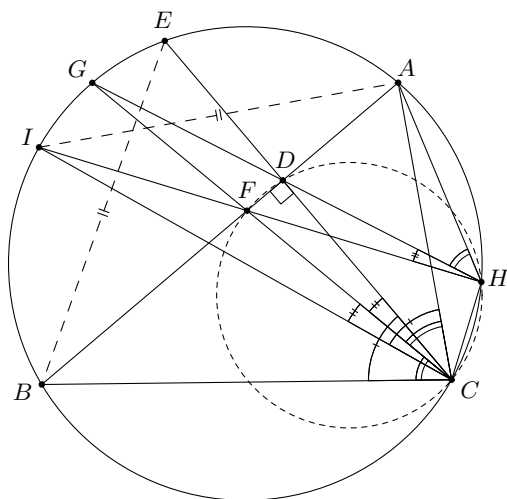


Figure 1

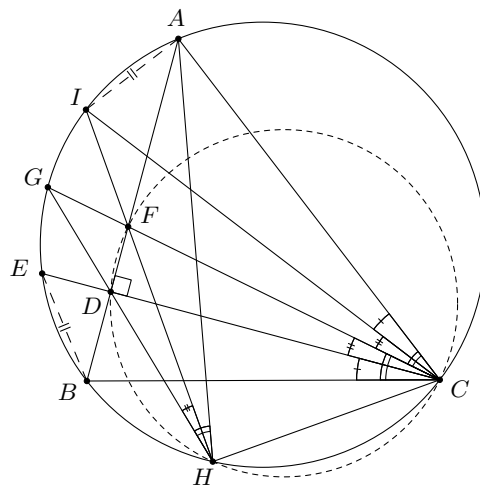


Figure 2

Solution. Since CG bisects $\angle ACB$, we have $\angle AHG = \angle ACG = \angle GCB$. If the perpendicular to AB from C is between the angle bisector from C and CA (see Figure 1), then $\angle HDB = \angle HAB + \angle AHG = 180^\circ - \angle HCB + \angle GCB = 180^\circ - \angle GCH$. In the other case (see Figure 2), we find that $\angle HDB = \angle HAB + \angle AHG = \angle HCB + \angle GCB = \angle GCH$. In both cases, $CFDH$ is a cyclic quadrilateral. Thus, $\angle GCE = \angle FCD = \angle FHD = \angle IHG = \angle ICG$. In view of $\angle ACG = \angle GCB$ we obtain $\angle ACI = \angle ECB$, which implies $AI = EB$. $\square$

Problem 15. What is the least possible number of cells that can be marked on an $n \times n$ board such that for each $m > n/2$ both diagonals of any $m \times m$ sub-board contain a marked cell?

Solution. For any n it is possible to set n marks on the board and get the desired property, if they are simply put on every cell in row number $\lceil \frac{n}{2} \rceil$. We now show that n is also the minimum amount of marks needed.

If n is odd, then there are $2n$ series of diagonal cells of length $> \frac{n}{2}$ and both end cells on the edge of the board, and, since every mark on the board can at most lie on two of these diagonals, it is necessary to set at least n marks to have a mark on every one of them.

If n is even, then there are $2n-2$ series of diagonal cells of length $> \frac{n}{2}$ and both end cells on the edge of the board. We call one of these diagonals even if every coordinate (x, y) on it satisfies $2 \mid x-y$ and odd else. It can be easily seen that this is well-defined. Now, by symmetry, the number of odd and even diagonals is the same, so there are exactly $n-1$ of each of them. Any mark set on the board can at most sit on two diagonals and these two have to be of the same kind. Thus, we will need at least $\frac{n}{2}$ marks for the even diagonals, since there are $n-1$ of them and $2 \nmid n-1$, and, similarly, we need at least $\frac{n}{2}$ marks for the odd diagonals. So we need at least $\frac{n}{2} + \frac{n}{2} = n$ marks to get the desired property. $\square$

Problem 16. Let $\mathbb{R}$ denote the set of real numbers. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x^2) + f(xy) = f(x)f(y) + yf(x) + xf(x+y), \quad \text{for all } x, y \in \mathbb{R}.$$

Solution. Setting $x = 0$ in the equation we get $f(0)f(y) = (2 - y)f(0)$. If $f(0) \neq 0$, then $f(y) = 2 - y$ and it is easy to verify that this is a solution to the equation. Now assume $f(0) = 0$. Setting $y = 0$ in the equation we get $f(x^2) = xf(x)$. Interchanging x and y and subtracting from the original equation we get

$$xf(x) - yf(y) = yf(x) - xf(y) + (x - y)f(x + y)$$

or equivalently,

$$(x - y)(f(x) + f(y)) = (x - y)f(x + y).$$

For $x \neq y$ we therefore have $f(x + y) = f(x) + f(y)$. Since $f(0) = 0$ this clearly also holds for $x = 0$, and for $x = y \neq 0$ we have

$$f(2x) = f\left(\frac{x}{3}\right) + f\left(\frac{5x}{3}\right) = f\left(\frac{x}{3}\right) + f\left(\frac{2x}{3}\right) + f(x) = f(x) + f(x)$$

Setting $x = y$ in the original equation, using $f(x^2) = xf(x)$ and $f(2x) = 2f(x)$ we get

$$0 = f(x)^2 + xf(x) = f(x)(f(x) + x)$$

So for each x , either $f(x) = 0$ or $f(x) = -x$. But then

$$f(x) + f(y) = f(x + y) = \begin{cases} 0, & \text{or} \\ -(x + y), & \text{otherwise.} \end{cases}$$

and we conclude that $f(x) = -x$ if and only if $f(y) = -y$ when $x, y \neq 0$. We therefore have either $f(x) = -x$ for all x or $f(x) = 0$ for all x . It is easy to verify that both are solutions to the original equation. $\square$